

SHAIBI SHAMSUDEEN

AI SYSTEMS ARCHITECT | ENGINEERING LEADER | AI SAFETY RESEARCHER

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PROFESSIONAL SUMMARY

AI systems and engineering leader with 10+ years of experience across intelligent applications, data and machine learning systems, workflow automation, governance, and enterprise delivery. Architected multi-agent research, conversational AI, real-time mapping, risk-assessment, and algorithmic-trading solutions from requirements through deployment. Leads a five-person engineering and quantitative team and translates business, compliance, and reliability requirements into modular technical designs. Co-author of research on LLM safety, mechanistic interpretability, and inference-time refusal control.

CORE EXPERTISE

AI Architecture: Generative AI, LLM applications, agentic workflows, multi-agent orchestration, RAG, semantic search, function calling, prompt design

AI Reliability: Model evaluation, backtesting, AI safety, refusal behavior, activation steering, observability, cost and performance trade-offs

Engineering: Python, TypeScript, SQL, React, Flask, FastAPI, REST APIs, event-driven systems, data pipelines, system integration

Data and AI: pandas, NumPy, scikit-learn, TensorFlow, PyTorch, Matplotlib, LangGraph, LangChain, OpenAI, Gemini, Hugging Face, ChromaDB

Platforms and Delivery: PostgreSQL, MySQL, Supabase, AWS, Docker, Git, GitHub, CI/CD, Vercel, Power BI, Tableau, Streamlit

Leadership: Solution architecture, requirements engineering, cross-functional delivery, Agile/Scrum, stakeholder management, GRC and auditability

PROFESSIONAL EXPERIENCE

Head of Engineering Unit | Proceedit Trading | Barcelona, Spain (Remote from Dubai) | Jan 2024 - Present

- Supervise data-science engineering across three quantitative analysts and two software engineers building AI-driven trading engines, analytics services, and automation bots.
- Own solution architecture and delivery from business requirements and data design through model development, API integration, deployment, and performance review.
- Architected Python, TensorFlow, PostgreSQL, and Flask pipelines for market-data ingestion, price prediction, correction, forecasting, and trade analytics.
- Personally designed a new combination algorithm within the correction engine, a post-processing layer applied after stock-price prediction and before corrected prices feed order and signal calculations; reduced error to below 1%.

AI Researcher | Algoverse AI Research | California, USA (Remote) | Apr 2025 - Present

- Co-author research on interpretable LLM refusal behavior using Llama 3 8B residual-stream activations, category-specific steering vectors, and inference-time control.
- Contributed to evaluation of safety and over-refusal behavior across benign, ambiguous, and harmful prompt sets, with emphasis on controllability and deployment reliability.
- Co-authored LLM-safety research accepted at the NeurIPS 2025 Mechanistic Interpretability Workshop and the COLM 2026 Actionable Interpretability Workshop.

Data Analyst | Avenir Arabia IT DMCC | Dubai, UAE | May 2023 - Dec 2023

- Translated client process problems into analytics requirements; cleaned and consolidated multi-source data and built Python, Power BI, and Tableau dashboards for decision support.
- Defined KPIs with business teams, quantified automation opportunities, and communicated findings to senior stakeholders.

Head of Product Development and Compliance Services | Tadashie FZCO | Dubai, UAE | Jan 2021 - Apr 2023

- Led functional and product design for GRC applications covering due diligence, conflicts of interest, anti-bribery controls, document management, approvals, and audit evidence.
- Integrated Python dashboards and API-connected workflows, improving process efficiency by 40% and increasing traceability of compliance decisions.

Senior Compliance Advisor and Functional Systems Lead | Petrofac International UAE Ltd. | Sharjah, UAE | Oct 2013 - Apr 2020

- Led functional architecture and product requirements for the Cerico third-party risk assessment platform, defining risk models, due-diligence questionnaires, workflow logic, approval controls, and audit evidence; Dow Jones acquired Cerico in 2018.
- Designed and evolved global compliance-platform workflows for policies, e-learning, gifts and hospitality, conflicts of interest, annual certifications, and risk-based due diligence, including ERP integration.
- Led compliance risk assessments, control reviews, investigations, policy implementation, and stakeholder adoption across multinational operations.

Quality Assurance Coordinator and Technical Assistant | Petrofac International UAE Ltd. | Sharjah, UAE | Jan 2008 - Oct 2013

- Designed an internal audit planning and findings-analysis tool, supported risk-based audit programs, and automated inspection coordination, reducing inspection costs by 30%.
- Managed quality systems, KPI reporting, lessons learned, and engineering and procurement assurance across major FEED and EPC projects.

SELECTED AI AND SYSTEMS PROJECTS

Luminar Multi-Agent AI Deep Researcher | Python, LangGraph, LangChain, ChromaDB, Streamlit | 2025

- Architected a modular multi-agent research system that routes work across web, video, academic, news, and vector-retrieval agents; added source consolidation, contradiction detection, caching, token and cost tracking, and PDF, DOCX, Markdown, and JSON export.
- Improved information-retrieval efficiency by 45% against the project baseline and won Second Prize among 250+ participants in the Outskill AI Engineering Accelerator.

AI Systems for Real Estate Intelligence | React, TypeScript, Gemini Live, Google Maps, Supabase | 2025 - 2026

- Architected real-time voice and text property discovery with model tool calling, Maps grounding, client profiles, persistent favorites, photorealistic 3D interaction, reusable content, analytics, and conversion workflows.
- Established an end-to-end chat-to-lead workflow with structured lead registration, CRM lifecycle management, real-time reporting, AI-assisted receipt scanning, expense registration, milestone-based commission invoicing, and tax invoices designed for UAE VAT requirements; built and deployed for AIN Global Real Estate and Lockwood & Carter.

Platform-Level Child AI Safety and Regulatory Assurance | Independent Product Research | 2026 - Present

- Developing the functional architecture for an operator-side age-assurance control and evidence layer that converts verified age signals and policy rules into executed controls, audit evidence, and continuous assurance; currently in competitive-gap and customer-validation stage.

RESEARCH PUBLICATIONS

- From Refusal Tokens to Refusal Control: Discovering and Steering Category-Specific Refusal Directions. Co-author. Accepted, COLM 2026 Actionable Interpretability Workshop; arXiv:2603.13359.
- What Do Refusal Tokens Learn? Fine-Grained Representations and Evidence for Downstream Steering. Co-author. Accepted poster, NeurIPS 2025 Mechanistic Interpretability Workshop.

EDUCATION AND PROFESSIONAL DEVELOPMENT

Master's Degree in Data Science | College de Paris | 2024

Master of Science in International Management | University of Liverpool | 2013

Bachelor of Science in Hospitality and Administration | IHMCT, India | 2004

- AI Engineering Accelerator, Outskill, 2025; Second Prize project award.
- Outskill Alumni continuing development, 2025 - Present: 40+ hours of focused labs in OpenClaw, Hermes Agent, Claude Code and Antigravity, Windsurf and Devin, open-source AI, production reliability, AI evaluations, delegation, and problem selection.
- Scrum Fundamentals Certified | IBM Data Analyst Capstone | KPMG Australia Data Analytics Consulting Virtual Internship | CCEP-I | ISO 37001 and ISO 9001 Lead Auditor training.

Earlier experience: Damac Properties, Unigreen Shipping and Air Freight, and Convergys India Services, 2004 - 2008.